Apache Pig — Detailed Explanation

Apache Pig is a high-level data processing platform that is mainly used to process and analyze large amounts of data in the Hadoop ecosystem.

To understand why Pig is needed, imagine a company has millions or billions of records. For example, an online shopping company may have data about customers, products, orders, payments, and website activity. Processing such a huge amount of data using normal programming can be difficult. Hadoop provides a way to process this large data, but traditionally we may need to write MapReduce programs, which can be lengthy and complicated. Apache Pig makes this process easier.

Instead of writing a complex MapReduce program, we can write simple instructions using a language called **Pig Latin**.

What is Pig Latin?

Pig Latin is the scripting language used by Apache Pig. It contains simple commands that describe what we want to do with the data.

LOAD → load data
FILTER → select required records
GROUP → group records
JOIN → combine datasets
ORDER → sort data
STORE → save the result

Example: Student Data

Suppose a college has a huge student dataset:

```
Name,Branch,Marks
Asha,CSE,85
Ravi,ISE,55
Manu,CSE,75
Priya,AI&ML,90
Kiran,ISE,40
```

Suppose the college has millions of student records and we want to find students who scored more than 60. In Pig, we can write:

```
students = LOAD 'students.csv'
USING PigStorage(',')
AS (name:chararray, branch:chararray, marks:int);

passed = FILTER students BY marks > 60;

STORE passed INTO 'output';
```

Step 1 — LOAD

LOAD reads the data from the file. Apache Pig takes the student data and loads it for processing.

Step 2 — FILTER

FILTER selects records based on a condition. Here, Pig selects students whose marks are greater than 60.

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students = LOAD 'students.csv'
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AS (name:chararray, branch:chararray, marks:int);

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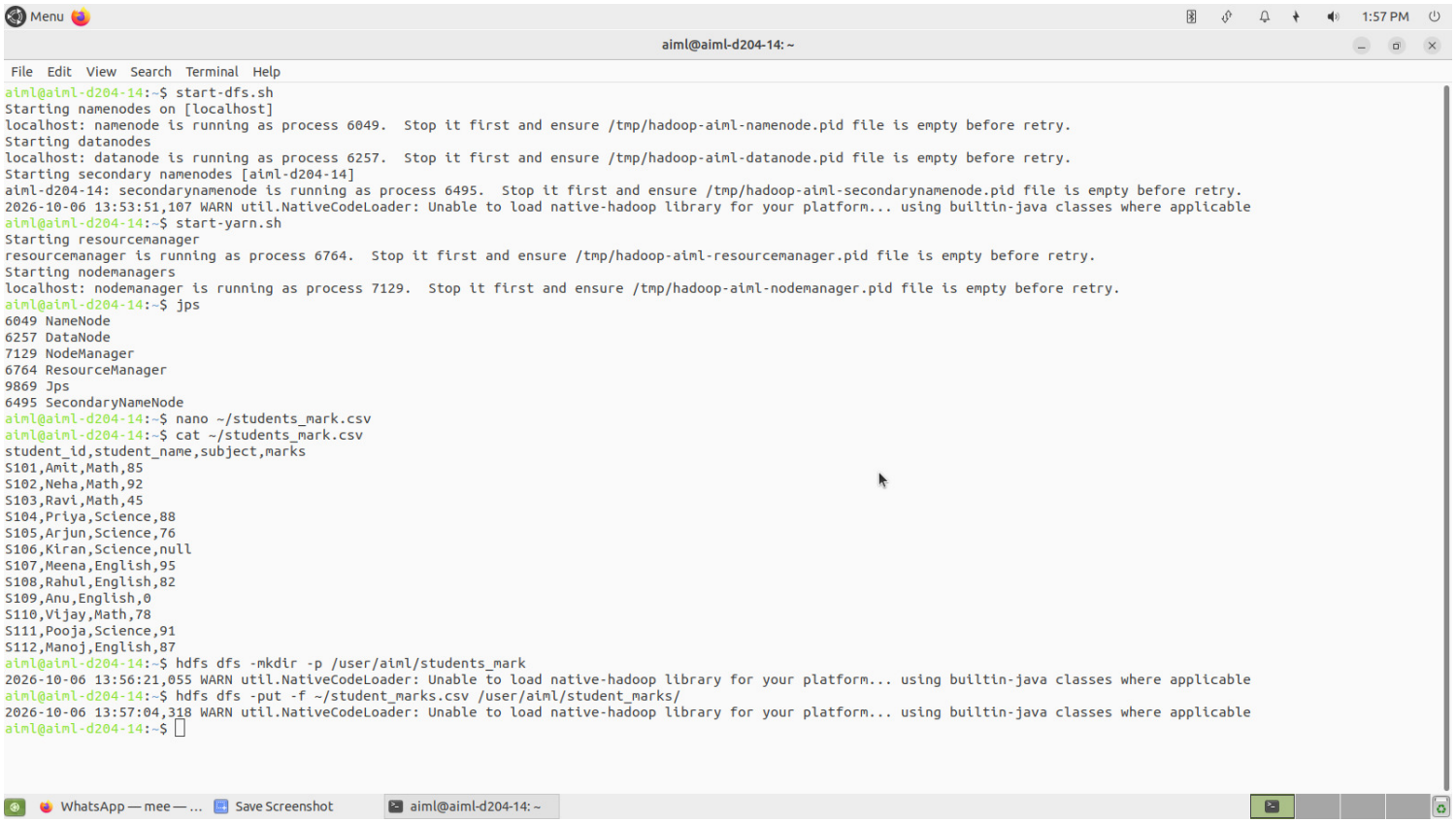
STORE passed INTO 'output';
```

Step 1 — LOAD

LOAD reads the data from the file. Apache Pig takes the student data and loads it for processing.

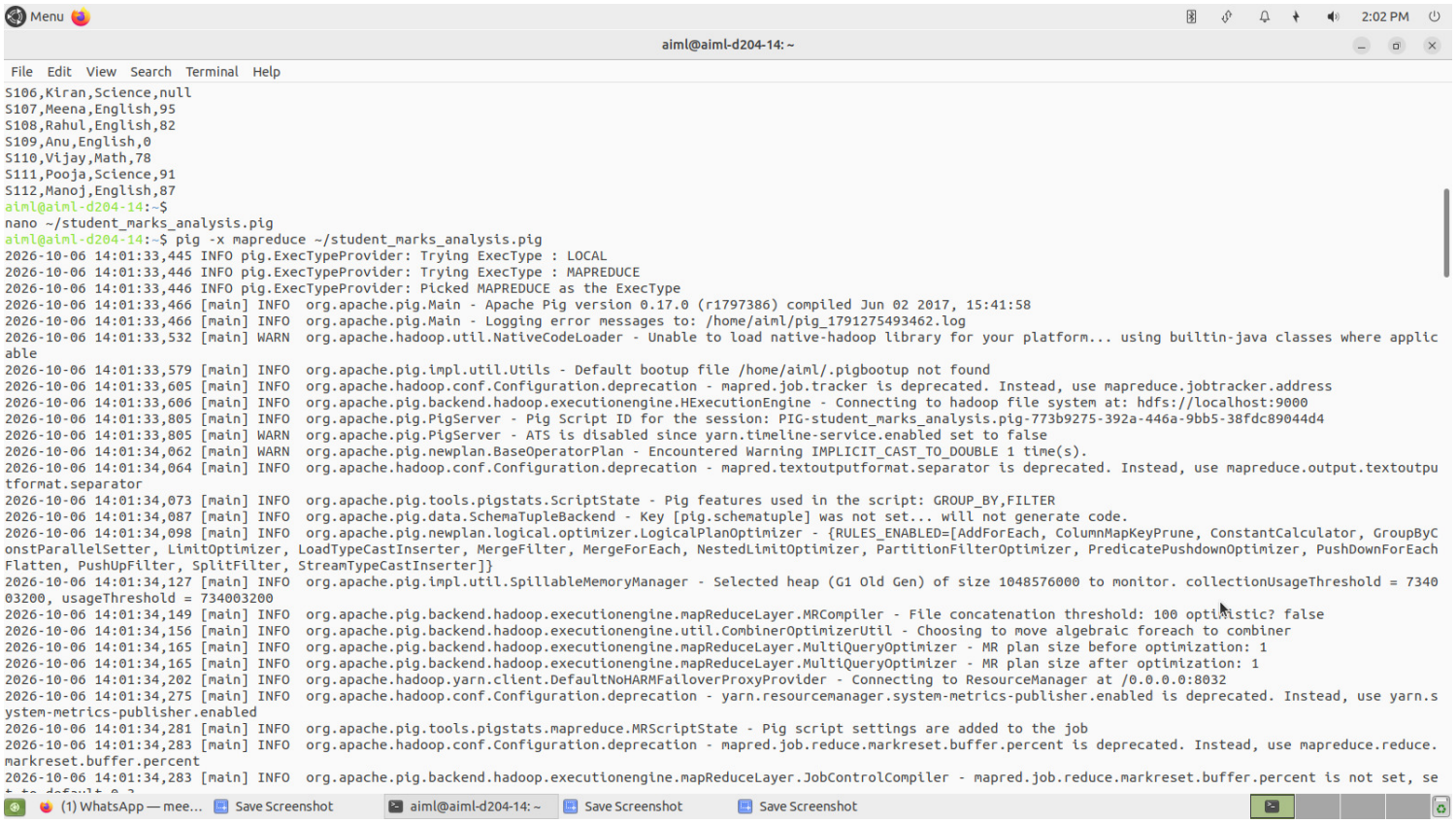
Step 2 — FILTER

FILTER selects records based on a condition. Here, Pig selects students whose marks are greater than 60.


Menu
1:57 PM
aiml@aiml-d204-14: ~
File Edit View Search Terminal Help
aiml@aiml-d204-14:~$ start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 6049. Stop it first and ensure /tmp/hadoop-aiml-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 6257. Stop it first and ensure /tmp/hadoop-aiml-datanode.pid file is empty before retry.
Starting secondary namenodes [aiml-d204-14]
aiml-d204-14: secondarynamenode is running as process 6495. Stop it first and ensure /tmp/hadoop-aiml-secondarynamenode.pid file is empty before retry.
2026-10-06 13:53:51,107 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 6764. Stop it first and ensure /tmp/hadoop-aiml-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 7129. Stop it first and ensure /tmp/hadoop-aiml-nodemanager.pid file is empty before retry.
aiml@aiml-d204-14:~$ jps
6049 NameNode
6257 DataNode
7129 NodeManager
6764 ResourceManager
9869 Jps
6495 SecondaryNameNode
aiml@aiml-d204-14:~$ nano ~/students_mark.csv
aiml@aiml-d204-14:~$ cat ~/students_mark.csv
student_id,student_name,subject,marks
S101,Amit,Math,85
S102,Neha,Math,92
S103,Ravi,Math,45
S104,Priya,Science,88
S105,Arjun,Science,76
S106,Kiran,Science,null
S107,Meena,English,95
S108,Rahul,English,82
S109,Anu,English,0
S110,Vijay,Math,78
S111,Pooja,Science,91
S112,Manoj,English,87
aiml@aiml-d204-14:~$ hdfs dfs -mkdir -p /user/aiml/students_mark
2026-10-06 13:56:21,055 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$ hdfs dfs -put -f ~/student_marks.csv /user/aiml/student_marks/
2026-10-06 13:57:04,318 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$
WhatsApp — mee — ...
Save Screenshot
aiml@aiml-d204-14: ~


Menu
2:00 PM
aiml@aiml-d204-14: ~
File Edit View Search Terminal Help
6049 NameNode
6257 DataNode
7129 NodeManager
6764 ResourceManager
9869 Jps
6495 SecondaryNameNode
aiml@aiml-d204-14:~$ nano ~/students_mark.csv
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S103,Ravi,Math,45
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S105,Arjun,Science,76
S106,Kiran,Science,null
S107,Meena,English,95
S108,Rahul,English,82
S109,Anu,English,0
S110,Vijay,Math,78
S111,Pooja,Science,91
S112,Manoj,English,87
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2026-10-06 13:56:21,055 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$ hdfs dfs -put -f ~/student_marks.csv /user/aiml/student_marks/
2026-10-06 13:57:04,318 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
aiml@aiml-d204-14:~$ hdfs dfs -cat /user/aiml/student_marks/student_marks.csv
2026-10-06 13:58:32,880 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
student_id,student_name,subject,marks
S101,Amit,Math,85
S102,Neha,Math,92
S103,Ravi,Math,45
S104,Priya,Science,88
S105,Arjun,Science,76
S106,Kiran,Science,null
S107,Meena,English,95
S108,Rahul,English,82
S109,Anu,English,0
S110,Vijay,Math,78
S111,Pooja,Science,91
S112,Manoj,English,87
aiml@aiml-d204-14:~$
nano ~/student_marks_analysis.pig
aiml@aiml-d204-14:~$
(1) WhatsApp — mee...
Save Screenshot
aiml@aiml-d204-14: ~
Save Screenshot
Save Screenshot


Menu
1:59 PM
aiml@aiml-d204-14: ~
File Edit View Search Terminal Help
GNU nano 6.2 /home/aiml/student_marks_analysis.pig
marks = LOAD '/user/aiml/student_marks/student_marks.csv'
USING PigStorage(',')
AS (
student_id:chararray,
student_name:chararray,
subject:chararray,
marks:chararray
);
data = FILTER marks BY student_id != 'student_id';
non_null = FILTER data BY marks IS NOT NULL AND marks != 'null';
typed_data = FOREACH non_null GENERATE
student_id,
student_name,
subject,
(float)marks AS marks;
valid_marks = FILTER typed_data BY marks >= 1.0;
grouped_marks = GROUP valid_marks BY subject;
average_marks = FOREACH grouped_marks GENERATE
group AS subject,
AVG(valid_marks.marks) AS average_marks;
STORE average_marks INTO '/user/aiml/student_marks/output'
USING PigStorage(',');
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next
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Save Screenshot
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Menu
2:02 PM
aiml@aiml-d204-14: ~
File Edit View Search Terminal Help
S106,Kiran,Science,null
S107,Meena,English,95
S108,Rahul,English,82
S109,Anu,English,0
S110,Vijay,Math,78
S111,Pooja,Science,91
S112,Manoj,English,87
aiml@aiml-d204-14:~$
nano ~/student_marks_analysis.pig
aiml@aiml-d204-14:~$ pig -x mapreduce ~/student_marks_analysis.pig
2026-10-06 14:01:33,445 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-06 14:01:33,446 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-06 14:01:33,446 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-06 14:01:33,466 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-10-06 14:01:33,466 [main] INFO org.apache.pig.Main - Logging error messages to: /home/aiml/pig_1791275493462.log
2026-10-06 14:01:33,532 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-10-06 14:01:33,579 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file /home/aiml/.pigbootup not found
2026-10-06 14:01:33,605 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-10-06 14:01:33,606 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-10-06 14:01:33,805 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-student_marks_analysis.pig-773b9275-392a-446a-9bb5-38fdc89044d4
2026-10-06 14:01:33,805 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-10-06 14:01:34,062 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
2026-10-06 14:01:34,064 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-10-06 14:01:34,073 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-10-06 14:01:34,087 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-10-06 14:01:34,098 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-10-06 14:01:34,127 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, usageThreshold = 734003200
2026-10-06 14:01:34,149 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-10-06 14:01:34,156 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-10-06 14:01:34,165 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-10-06 14:01:34,165 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-10-06 14:01:34,202 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-10-06 14:01:34,275 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-10-06 14:01:34,281 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-10-06 14:01:34,283 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-10-06 14:01:34,283 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, se
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2:04 PM
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File Edit View Search Terminal Help
2026-10-06 14:02:51,355 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-10-06 14:02:51,359 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-10-06 14:02:52,360 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:53,361 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:54,362 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:55,363 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:56,364 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:57,365 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:58,366 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:02:59,367 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:03:00,368 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:03:01,368 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:03:01,472 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-10-06 14:03:02,473 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-10-06 14:03:03,474 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
^Z
[1]+ Stopped pig -x mapreduce ~/student_marks_analysis.pig
aiml@aiml-d204-14:~$ hdfs dfs -ls /user/aiml/student_marks/output
2026-10-06 14:03:32,874 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 aiml supergroup 0 2026-10-06 14:01 /user/aiml/student_marks/output/_SUCCESS
-rw-r--r-- 1 aiml supergroup 36 2026-10-06 14:01 /user/aiml/student_marks/output/part-r-00000
aiml@aiml-d204-14:~$
hdfs dfs -cat /user/aiml/student_marks/output/part-r-00000
2026-10-06 14:04:00,576 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Math,75.0
English,88.0
Science,85.0
aiml@aiml-d204-14:~$
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Save Screenshot
aiml@aiml-d204-14: ~
Save Screenshot
Save Screenshot
Save Screenshot